

ECN 594: Midterm Exam

February 9, 2026

Instructions:

- You have **80 minutes**
- You may bring a calculator and notes on a two-sided cheat-sheet (letter-size paper)
- Please be neat. If your work is too messy it will not be graded.
- Be sure to show your working.
- This is a long exam, so there are lots of ways to get points. If you get stuck, move on!
- Good luck!

Name: _____

Question:	1	2	3	4	Total
Points:	30	30	20	20	100
Score:					

1. Short Answer Questions (30 points)

1. For each question, give a number, True/False/NEI, or a 1-2 sentence answer.
 - (a) (4 points) A monopolist faces demand with elasticity $\varepsilon = -2$ and has $MC = \$30$. What is the profit-maximizing price?
 - (b) (4 points) Consider the logit utility: $u_{ijt} = \beta \cdot \text{sugar}_j + \alpha \cdot p_{jt} + \xi_{jt} + \varepsilon_{ijt}$. What is the economic interpretation of α ?
 - (c) (4 points) True, False, or NEI: If we add 5 to the mean utility of all goods (including the outside option), the market shares do not change.
 - (d) (4 points) Why are BLP instruments (characteristics of other products) *relevant* for price? (1-2 sentences)
 - (e) (4 points) True, False, or NEI: In the logit model with small market shares, higher-priced products typically have more elastic demand.
 - (f) (3 points) True, False, or NEI: Under average cost pricing ($P = AC$), deadweight loss is zero.
 - (g) (4 points) A researcher claims that consumer surplus in logit is $CS = \frac{1}{|\alpha|} (\exp(\delta_1) + \exp(\delta_2))$. Consider a market with two goods ($\delta_1 = \delta_2 = 2$) and an outside option. Is this formula correct? Explain.

(h) (3 points) True, False, or NEI: Under perfect price discrimination, deadweight loss is zero.

2. Demand Estimation (30 points)

2. A researcher wants to estimate demand for automobiles using a logit model:

$$u_{ijt} = \beta \cdot \text{mpg}_j + \alpha \cdot p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

where mpg_j is fuel efficiency, p_{jt} is price, ξ_{jt} is unobserved product quality, and ε_{ijt} is i.i.d. Type 1 Extreme Value.

- (a) (5 points) Define the mean utility δ_{jt} . Using the Berry inversion, derive an expression for δ_{jt} in terms of observed market shares s_{jt} and the outside option share s_{0t} .
- (b) (5 points) Write down the linear regression equation the researcher would estimate to recover β and α . Clearly identify the dependent variable, regressors, and error term.
- (c) (5 points) Why does OLS produce biased estimates of α ? What is the likely direction of the bias? (2-3 sentences)
- (d) (5 points) A researcher proposes using “Hausman instruments” (prices of the same car in other markets) to address the endogeneity problem. Explain one reason why these instruments might be problematic.

- (e) (5 points) Compared to the basic logit, what is one advantage of adding demographic interactions (e.g., $\text{income} \times \text{price}$) to the model?
- (f) (5 points) A researcher takes all Toyota Corollas and splits them into two “products” based on the last digit of their license plate: “Corolla-Odd” and “Corolla-Even.” What will the logit model predict about the change in consumer surplus from this split? Should we believe this prediction? Explain.

3. Two-Part Tariffs (20 points)

3. A gym charges a monthly membership fee F and a per-visit price p . Marginal cost per visit is $c = 2$.

Part A: Homogeneous consumers

There are 100 identical consumers, each with demand $q = 20 - 2p$ for gym visits per month.

- (a) (5 points) Write down the gym's profit as a function of F and p . (Hint: What is consumer surplus at price p ? What is the maximum F the gym can charge?)

- (b) (5 points) Show that the profit-maximizing per-visit price is $p^* = c = 2$. (Hint: Take the derivative of profit with respect to p and set it equal to zero.)

Part B: Heterogeneous consumers

Now suppose there are 50 "heavy users" with demand $q_H = 10 - p$ and 100 "light users" with demand $q_L = 5 - p$. Marginal cost is $c = 0$.

- (c) (10 points) The gym is considering two pricing strategies:

- (a) $F = 50, p = 0$
- (b) $F = 12.5, p = 0$

Calculate profit under each strategy. Which is better and why?

4. Self-Selection (20 points)

4. A software company sells two versions: Pro (P) and Basic (B). There are two types of users: businesses with high willingness to pay and students with low willingness to pay.

Consumer Type	Willingness to Pay	
	Pro Version	Basic Version
Business (40 users)	\$400	\$150
Student (60 users)	\$120	\$100

Marginal cost is \$30 for both versions.

- (a) (4 points) If the company could perfectly identify consumer types, what prices would it charge for each version to each type? What is total profit?
- (b) (4 points) The company cannot identify types but offers both versions. Write down the incentive compatibility (IC) constraint that ensures businesses buy Pro rather than Basic.
- (c) (6 points) Suppose the company sets $P_P = \$380$ and $P_B = \$100$. Explain why these prices are not optimal. Which constraint is violated?
- (d) (6 points) Find the profit-maximizing prices for Pro and Basic under self-selection. (Hint: Which IC constraint binds? Set it as an equality and solve.)